



Generating a sequence using a term-to-term rule

Task A: Using term-to-term rules

- 1) This crossnumber is made from sequences written from left to right or from top to bottom. Fill in the crossnumber according to the given rules. Fill in the missing rule for 'A: down'

	A -4					
B 7	1	-5	E -11	-17	F -23	
	6		6		6	
	11		23		35	
C 4	16	28	40	52	64	76
	21				93	
	D 26	50	74	98	122	

Across

- B: add -6 each time
C: add 12 each time
D: add 24 each time

Down

- A: add 5 each time
E: add 17 each time
F: add 29 each time

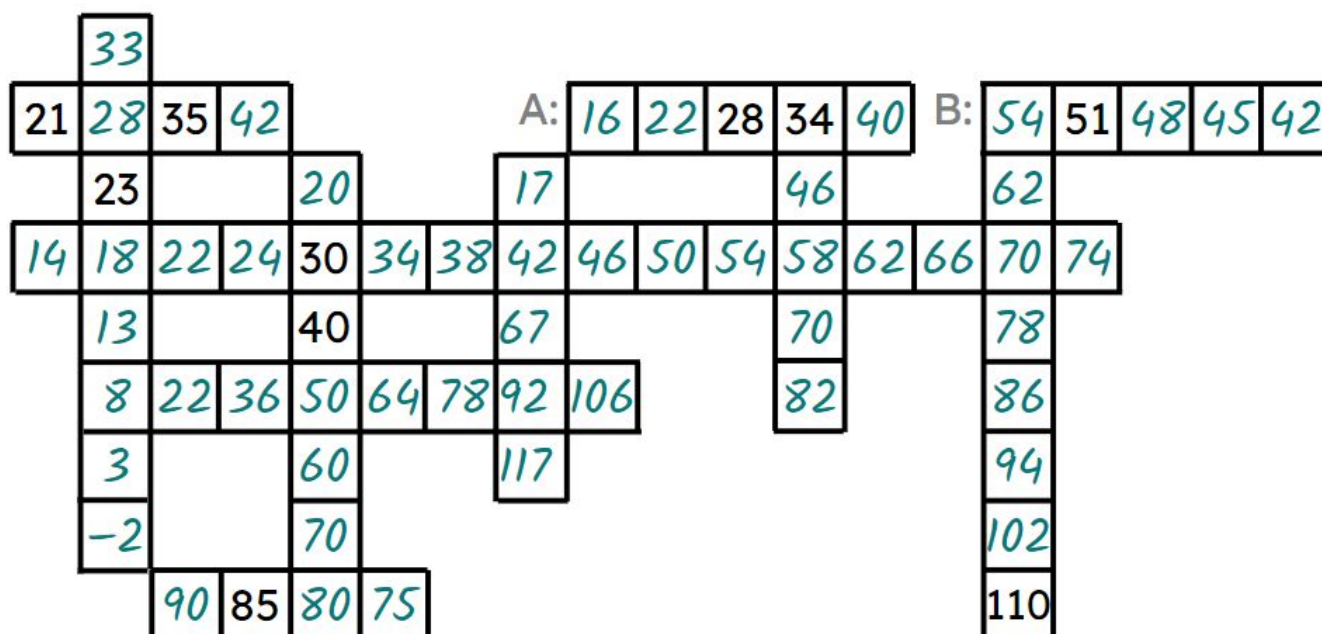
- 2) For each sequence, use the term to term rule to find the desired term.

- | | |
|--|-----|
| a) Start on 2 and add 7 each time. Find the value of T_3 | 16 |
| b) Start on 1 and multiply by 3 each time. Find the value of T_4 | 27 |
| c) Start on 6, double then add 10 each time. Find the value of T_4 | 111 |
| d) Start on 5, add -4 each time. Find the value of T_6 | -15 |
| e) Start on 10, multiply by 2 each time. Find the value of T_7 | 640 |
| f) Start on -4, add 5 each time. Find the value of T_{10} | 41 |
| g) Start on 24, add -2 each time. Find the value of T_{25} | -24 |



Task B: Finding term-to-term rules

- 1) Fill in each row and column so they form a sequence with an additive pattern. Then fill in the **term-to-term** rules for the stated rows.



A **term-to-term** rule for row A is 'Add 6'. A **term-to-term** rule for row B is 'Add -3' or 'Minus 3'.

- 2) Describe a **term-to-term** rule for each sequence below

a) 4, 7, 10, ...
E.g. 'add 3'

d) 5, -3, -11, ...
E.g. 'add -8'

b) 13, 26, 52, ...
E.g. 'multiply by 2'

e) 7, -21, 63, ...
E.g. 'multiply by -3'

c) 0.2, 2, 20, ...
E.g. 'multiply by 10'

f) 2, 5, 11, ...
E.g. 'multiply by 2 then add 1'
or 'add 3, add 6, add 9, ...'

- 3) Alex and Aisha are writing **term-to-term** rules for a sequence with $T_1 = 5$ and $T_2 = 15$

- a) Explain why Alex is incorrect

It is possible that multiple rules can apply to the same two terms.

- b) What rules could they have written?

E.g. 'add 10', 'Multiply by 3', 'Multiply by 2 then add 5' 'Add 10, then 20, then 30, ...'

Aisha and I have different rules so one of us must be wrong.



Alex

